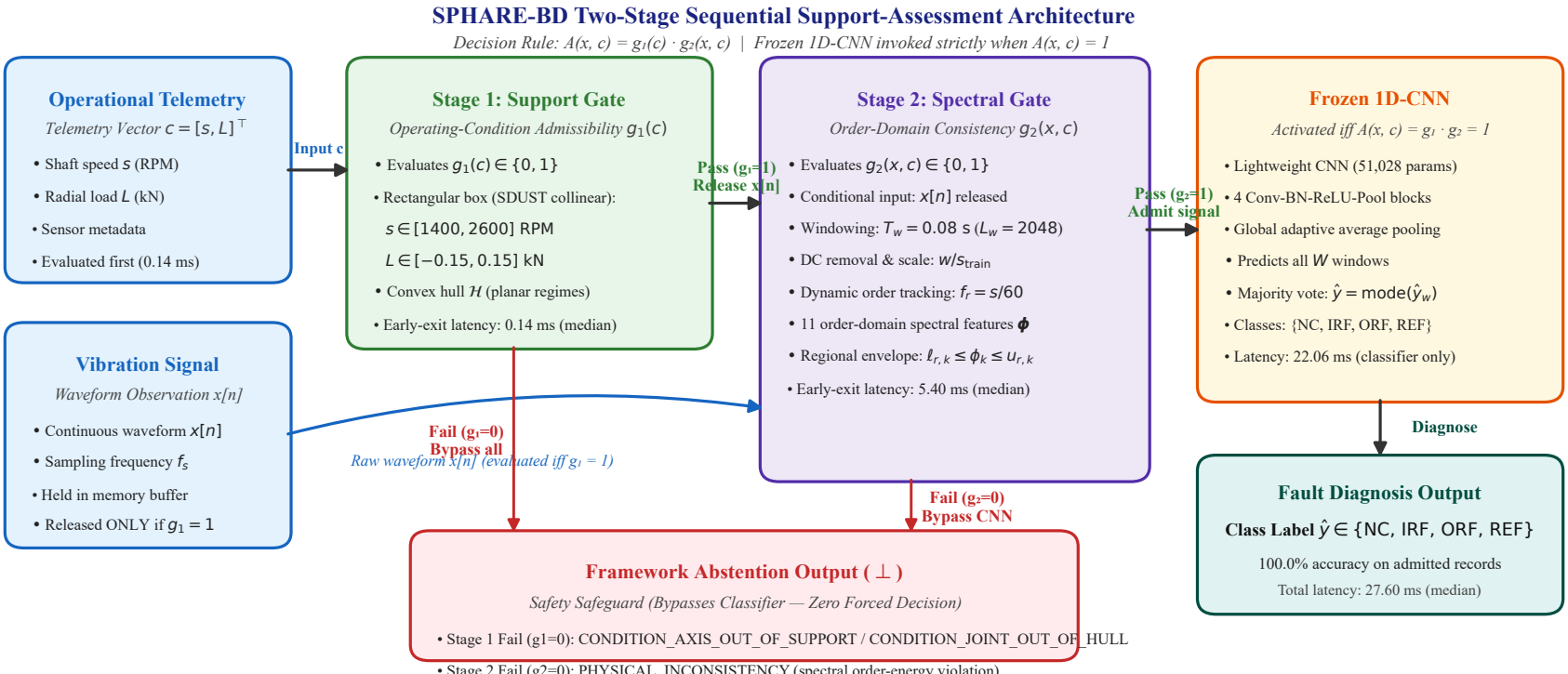

SPHARE-BD Two-Stage Sequential Support-Assessment Architecture
Decision Rule: A(x, c) = g₁(c) · g₂(x, c) | Frozen 1D-CNN invoked strictly when A(x, c) = 1
Operational Telemetry
Telemetry Vector c = [s, L]ᵀ
• Shaft speed s (RPM)
• Radial load L (kN)
• Sensor metadata
• Evaluated first (0.14 ms)
Input c
Vibration Signal
Waveform Observation x[n]
• Continuous waveform x[n]
• Sampling frequency fs
• Held in memory buffer
• Released ONLY if g1 = 1
Stage 1: Support Gate
Operating-Condition Admissibility g1(c)
• Evaluates g1(c) ∈ {0, 1}
• Rectangular box (SDUST collinear):
s ∈ [1400, 2600] RPM
L ∈ [−0.15, 0.15] kN
• Convex hull ℋ (planar regimes)
• Early-exit latency: 0.14 ms (median)
Pass (g1=1)
Release x[n]
Fail (g1=0)
Bypass all
Raw waveform x[n] (evaluated iff g1 = 1)
Stage 2: Spectral Gate
Order-Domain Consistency g2(x, c)
• Evaluates g2(x, c) ∈ {0, 1}
• Conditional input: x[n] released
• Windowing: Tw = 0.08 s (Lw = 2048)
• DC removal & scale: w/strain
• Dynamic order tracking: fr = s/60
• 11 order-domain spectral features ϕ
• Regional envelope: ℓr,k ≤ ϕk ≤ ur,k
• Early-exit latency: 5.40 ms (median)
Fail (g2=0)
Bypass CNN
Pass (g2=1)
Admit signal
Frozen 1D-CNN
Activated iff A(x, c) = g1 · g2 = 1
• Lightweight CNN (51,028 params)
• 4 Conv-BN-ReLU-Pool blocks
• Global adaptive average pooling
• Predicts all W windows
• Majority vote: ŷ = mode(ŷw)
• Classes: {NC, IRF, ORF, REF}
• Latency: 22.06 ms (classifier only)
Diagnose
Fault Diagnosis Output
Class Label ŷ ∈ {NC, IRF, ORF, REF}
100.0% accuracy on admitted records
Total latency: 27.60 ms (median)
Framework Abstention Output (⊥)
Safety Safeguard (Bypasses Classifier — Zero Forced Decision)
• Stage 1 Fail (g1=0): CONDITION_AXIS_OUT_OF_SUPPORT / CONDITION_JOINT_OUT_OF_HULL
• Stage 2 Fail (g2=0): PHYSICAL_INCONSISTENCY (spectral order-energy violation)